RESEARCH PAPER

CONTENT MATURITY, RELEASE LIFECYCLE AND PLAYLIST ROTATION ANALYSIS OF SPAIN TOP 50

A Music Analytics Study

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Project Domain: Data Analytics and Music Industry

Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly and Streamlit

GitHub: https://github.com/Vamshika2029/Atlantic-Spain-Top50-Analytics

ABSTRACT

Music streaming services and curated playlists influence how listeners discover songs and how tracks gain visibility. This project presents an analytical study of the Spain Top 50 playlist, focusing on content maturity, song lifecycle, playlist rotation and popularity patterns. A dataset containing chart observations, song and artist information, popularity scores, release characteristics and explicit-content indicators was analyzed using Python.

The study includes Exploratory Data Analysis, Song Lifecycle Analysis, Lifecycle Stage Classification, Playlist Churn Analysis, Content Maturity Analysis and Popularity Lifecycle Analysis. A Streamlit dashboard was developed to present the results through interactive visualizations. The analysis identified 575 unique song-artist combinations, an average observed chart duration of approximately 48.3 days and an average retention rate of 96.56% between consecutive available playlist snapshots. These results describe patterns in song longevity, playlist stability, release characteristics and popularity changes.

Keywords: Music Analytics, Spain Top 50, Song Lifecycle, Playlist Churn, Content Maturity, Popularity Analysis, Python, Streamlit.

1. INTRODUCTION

Digital streaming has transformed music consumption by making large catalogs and curated playlists widely accessible. Playlists can provide exposure to songs and change as new tracks gain attention or existing tracks lose visibility. Analyzing playlist snapshots can help describe song longevity, chart movement, playlist turnover and relationships between song characteristics and observed performance. This project examines the Spain Top 50 playlist using data analytics methods.

2. PROBLEM STATEMENT

Playlist composition changes over time, but these changes are not always easy to interpret from individual chart snapshots. It can be difficult to determine how long songs remain visible, how their positions evolve, how frequently tracks enter or leave, and whether release or content characteristics are associated with different observed outcomes. This project addresses these questions through structured analysis of the available dataset.

3. OBJECTIVES

• Perform exploratory analysis of Spain Top 50 playlist data.

• Measure song lifecycle duration, observed chart days and peak position.

• Classify observations into project-defined lifecycle stages.

• Measure playlist entries, exits, churn and retention.

• Compare explicit and clean content, and singles and album tracks.

• Examine popularity changes during observed chart appearances.

• Build an interactive Streamlit dashboard.

4. DATASET DESCRIPTION

The dataset contains dated playlist observations and song-level attributes. The fields include date, position, song, artist, popularity, duration_ms, album_type, total_tracks, is_explicit and album_cover_url. The analysis identified 575 unique song-artist combinations.

Attribute	Description
Date	Date of playlist observation
Position	Song's playlist rank
Song / Artist	Track title and performing artist

Attribute	Description
Popularity	Source-provided popularity score
Duration_ms	Song duration in milliseconds
Album_type	Single or Album
Total_tracks	Number of tracks in the release
Is_explicit	Explicit-content indicator
Album_cover_url	Album artwork reference

5. TOOLS AND TECHNOLOGIES

Python was the primary programming language. **Pandas** and **NumPy** supported data preparation and numerical operations. **Matplotlib** and **Seaborn** supported exploratory visualizations. **Plotly** provided interactive charts, while **Streamlit** was used to build the dashboard. **Visual Studio Code** was the development environment and **GitHub** was used for source-code version control and hosting.

6. RESEARCH METHODOLOGY

6.1 Data preparation

The dataset was inspected and prepared. Date values, numerical attributes and content categories were processed to support consistent calculations.

6.2 Exploratory analysis

Descriptive statistics and charts were used to inspect song duration, release type, explicit content, artist representation and daily playlist observations.

6.3 Song lifecycle

For each song-artist combination, first and last observed dates, observed chart days, calendar span, peak position and time to peak were calculated.

6.4 Lifecycle stages

Observations were classified into New Entry, Growth Phase, Peak Phase, Mature Phase and Decline Phase using entry age, chart position and comparisons with previous observed positions. These are project-defined labels, not official platform categories.

6.5 Playlist churn

Consecutive available snapshots were compared to identify new entries, exits and retained songs. Daily and monthly churn and retention measures were calculated.

6.6 Content maturity

Song-level records were grouped by explicit/clean classification and by single/album release type.

6.7 Popularity lifecycle

First, last, maximum and average popularity scores were examined, together with popularity change and correlation measures.

6.8 Dashboard

Results were presented through five Streamlit sections: Overview, Song Lifecycle, Playlist Rotation, Content Maturity and Popularity Analysis.

7. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis examined song distribution by release type and explicit-content classification, song duration, daily entries, artist representation and descriptive statistics. Visualizations were generated to identify patterns and provide context for subsequent analyses.

8. SONG LIFECYCLE ANALYSIS

Song lifecycle analysis measured how long each song appeared in the observed playlist data. Observed chart days count recorded appearances, while calendar span measures elapsed time between first and last observed dates. These measures differ because a song may have gaps between recorded appearances.

Metric	Result
Unique song-artist combinations	575
Average calendar span	57.44 days
Average observed chart days	48.26 days
Average time to peak	9.48 days
Average peak position	21.79

La Bachata by Manuel Turizo had the longest observed chart presence, with 538 observed days. Other long-running tracks included Columbia, VAGABUNDO, LA FALDA and LUNA.

9. LIFECYCLE STAGE CLASSIFICATION

Songs were classified into New Entry, Growth Phase, Peak Phase, Mature Phase and Decline Phase. The rules considered time since entry, chart position and comparisons with previous observed positions. Mature Phase represented the largest group in the analyzed results. These stages are heuristic categories developed for this project, not official streaming-platform classifications.

10. PLAYLIST CHURN ANALYSIS

Playlist churn measures changes between consecutive available snapshots. New entries are tracks present in the current snapshot but absent from the previous one; exits are tracks present in the previous snapshot but absent from the current one; retained tracks appear in both.

Metric	Result
Average new entries	1.72
Average exits	1.72
Average entry churn rate	3.44%
Average exit churn rate	3.44%
Average retention rate	96.56%
Average interval between snapshots	1.01 days

The results indicate relatively high retention between consecutive available snapshots. Equal average entries and exits are consistent with a playlist that generally contains 50 tracks. November 2025 had a higher churn rate in the available results, but it was a partial month ending November 27. Snapshot gaps should be considered when interpreting daily comparisons.

11. CONTENT MATURITY ANALYSIS

11.1 Explicit versus clean

Content type	Unique songs	Average popularity
Clean	325	71.13
Explicit	250	69.26

Clean songs had a higher average popularity score than explicit songs in this dataset. This observed difference does not establish that content classification causes popularity differences.

11.2 Singles versus albums

Release type	Unique songs	Average popularity
Album	292	65.84
Single	283	74.94

Singles had a higher average popularity score than album tracks in the analyzed data. This comparison describes the dataset and should not be generalized to all releases or markets.

11.3 Duration and album size

The correlation between song duration and observed chart days was approximately 0.015, indicating almost no linear relationship. The correlation between album size and observed chart days was approximately -0.153, indicating a weak negative relationship. Duration and album size alone were not strong linear indicators of chart longevity in this analysis.

12. POPULARITY LIFECYCLE ANALYSIS

Metric	Result
Songs analyzed	575
Average first popularity	55.99
Average last popularity	72.54
Average maximum popularity	74.26
Average popularity change	+16.55
Average days to peak popularity	17.03

Popularity increased for 409 songs, decreased for 45 songs and remained unchanged for 121 songs between first and last observed values. Correlations were: average popularity versus observed chart days, +0.371; average popularity versus peak chart position, -0.404; and observed chart days versus peak chart position, -0.509. Popularity is a source-provided score, not a direct measurement of actual streams or listener counts.

13. RESULTS AND DISCUSSION

• The analysis identified 575 unique song-artist combinations.

• Average observed chart duration was approximately 48.3 days.

• Average retention between consecutive available snapshots was 96.56%.

• Clean songs had a higher average popularity score than explicit songs in this dataset.

• Singles had a higher average popularity score than album tracks in this dataset.

• Popularity increased between first and last observed appearances for 409 songs.

• Average popularity had a positive association with observed chart duration.

• Song duration had almost no linear relationship with observed chart duration.

Together, these findings describe playlist stability, song performance and content characteristics in the analyzed Spain Top 50 data.

14. RECOMMENDATIONS

• Monitor new entries to identify emerging tracks and changes in playlist composition.

• Track lifecycle stages to summarize changing chart behavior.

• Review monthly churn patterns while accounting for partial months and snapshot gaps.

• Compare content categories without assuming causation.

• Examine single and album releases separately when reviewing performance.

• Monitor popularity scores alongside chart positions and observed longevity.

• Use interactive dashboards for regular performance reviews.

15. LIMITATIONS

• The study is limited to the available dataset and its observation period.

• Observed chart days count recorded appearances and may not equal continuous calendar availability.

• Lifecycle stages are project-defined heuristic labels, not official platform classifications.

• Popularity scores do not directly represent actual streams or listener counts.

• Correlation does not establish causation.

• Findings from one playlist should not automatically be generalized to all markets or platforms.

16. CONCLUSION

This project analyzed the Spain Top 50 playlist using data analytics techniques to understand content maturity, song lifecycle, playlist rotation and popularity patterns. The study identified 575 unique song-artist combinations, an average observed chart duration of approximately 48.3 days and an average retention rate of 96.56% between consecutive available snapshots.

Comparisons of explicit and clean songs, and singles and album tracks, provided additional insights into content characteristics and popularity patterns. Popularity lifecycle analysis found an average increase of 16.55 points between first and last observed appearances. An interactive Streamlit dashboard was developed to present the findings in an accessible format.

Overall, the project demonstrates how data analytics can be applied to playlist observations to describe song performance, playlist dynamics and content characteristics.

17. REFERENCES

• Spain Top 50 playlist dataset used for this project.

• Python Documentation: https://docs.python.org/3/

• Pandas Documentation: https://pandas.pydata.org/docs/

• NumPy Documentation: https://numpy.org/doc/

• Matplotlib Documentation: https://matplotlib.org/stable/

• Plotly Documentation: https://plotly.com/python/

• Streamlit Documentation: https://docs.streamlit.io/

• GitHub Documentation: https://docs.github.com/

18. PROJECT LINKS

GitHub Repository: https://github.com/Vamshika2029/Atlantic-Spain-Top50-Analytics

Streamlit Dashboard: Add the public deployment URL after deployment.

Project Explanation Video: Add the shareable video URL after recording and uploading.